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DATA 446

Gen AI Final Project

Abstract

The objective of this project is to develop a model that can both detect the emotional tone of a text message and generate emotionally appropriate responses. Modern AI systems are expected to produce responses that are not only coherent, but also emotionally aligned with the user. This project investigates a modular system consisting of an emotion classifier and a controllable response generator. By separating emotion understanding from response generation, the system aims to improve interpretability, controllability, and emotional alignment in conversational AI systems.

Introduction

Human conversation relies heavily on emotional understanding. In text-based communication, users expect responses that are both coherent and emotionally appropriate. While recent language models can generate fluent text, many systems still struggle to recognize emotional context and respond accordingly. Misinterpreting tone can lead to responses that appear insensitive or irrelevant. As conversational AI becomes more common in customer support, healthcare, and virtual assistants, emotionally aware dialogue generation has become increasingly important.

The problem addressed in this project is the development of a dialogue system that can both classify emotional tone and generate emotionally aligned responses. More specifically, the system should first determine the emotion expressed in a message and then generate responses that reflect an appropriate conversational tone. This combines two natural language processing tasks: emotion classification and controllable language generation.

Emotion detection in text has been studied using datasets such as GoEmotions and DailyDialog. GoEmotions contains comments labeled with fine-grained emotions, while DailyDialog provides conversational exchanges with emotion annotations.

These datasets are effective for emotion classification, but they focus more on emotion understanding than emotionally controlled response generation.

Prior work such as Affect-Driven Dialogue Generation [1] explores emotionally conditioned dialogue generation using end-to-end architectures. While effective, these approaches combine emotion detection and generation into a single model, making evaluation and interpretability more difficult. In contrast, this project uses a modular architecture that separates emotion classification and response generation for greater control and transparency.

The proposed system consists of two main components. First, an encoder-only transformer model will classify the emotional tone of an input message using GoEmotions. Second, a lightweight GPT-style model trained on DailyDialog will generate responses conditioned on both the message and the target emotion. The system will generate three candidate responses using top-k sampling and provide short explanations describing how each response aligns with the detected emotion.

Expected results include accurate emotion classification and coherent emotionally aligned responses. The project also aims to demonstrate that separating emotion understanding from generation improves interpretability and controllability in conversational AI systems.

Methodology

This project studies emotionally aware dialogue generation. The objective is to create a system capable of recognizing the emotional tone of a message and generating responses that match the emotional context of the conversation. The system combines two connected tasks: emotion classification and emotion-conditioned response generation.

The main challenge is maintaining both conversational coherence and emotional alignment. The methodology combines an encoder-only transformer for emotion classification and a GPT-style autoregressive transformer for response generation. The classifier learns contextual representations for emotion prediction, while the generator produces dialogue responses conditioned on emotional information.

The system consists of four stages:

1. Input Message: The system receives a user text message as input.

2. Emotion Classification Module: The first component is an encoder-only transformer trained on the GoEmotions dataset. The classifier predicts a fine-grained emotion label which is then mapped into broader DailyDialog emotion categories such as happy, sad, angry, neutral, etc.

3. Response Generation Module: The second component is a lightweight GPT-style language model trained on DailyDialog. The generator conditions on both the input message and the target emotion label. The model uses top-k sampling to generate three candidate responses with varied wording while maintaining emotional consistency.

4. Explanation Module:

The final component provides template-based explanations describing why each generated response aligns with the detected emotional tone. These explanations improve interpretability and transparency.

This project differs from previous work in several ways. Emotion classification and generation are separated into modular components. GoEmotions labels are mapped into conversational DailyDialog categories. The system generates multiple candidate responses for the user to select from using top-k sampling. Template-based explanations explain emotions and the responses to improve transparency and interpretability.

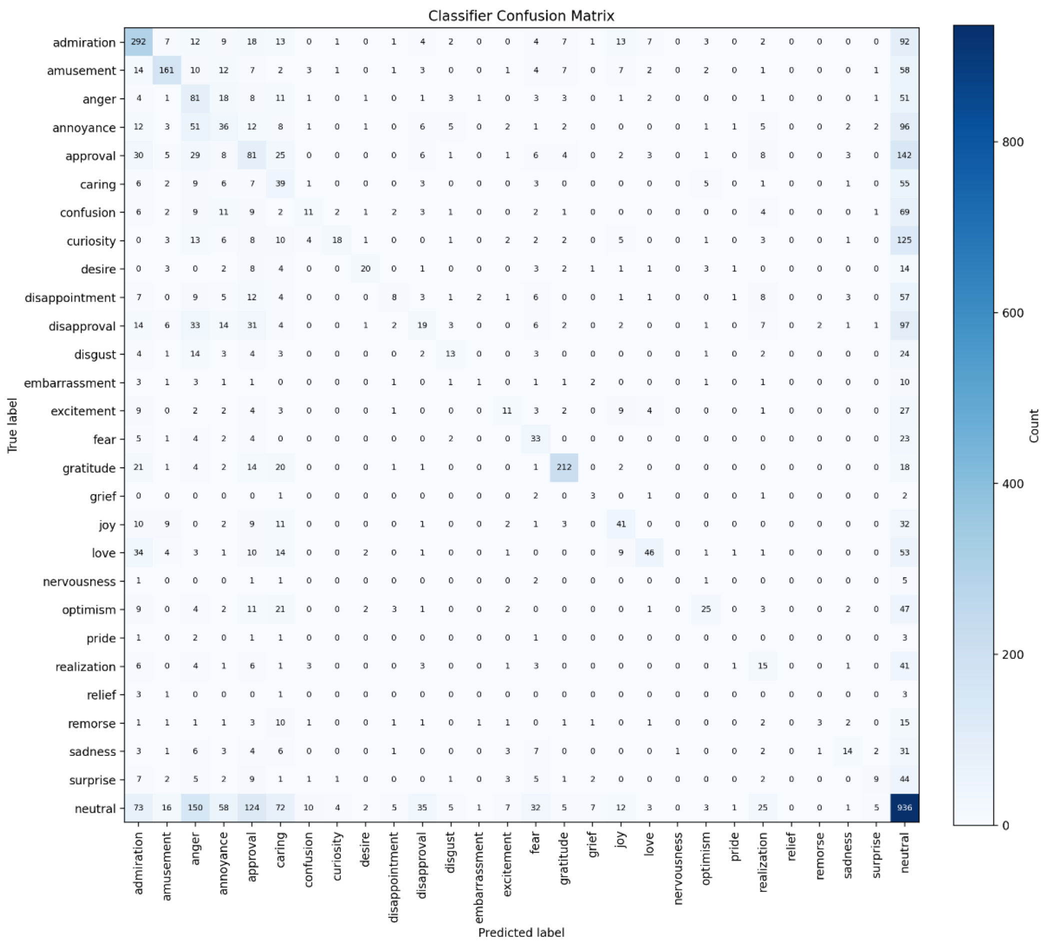
The emotion classifier will be trained on GoEmotions and evaluated on a held-out validation set. The classifier will be evaluated using accuracy, F1-score, and confusion matrices. The response generator will be trained on DailyDialog conversations. Generated responses will also be passed back through the emotion classifier to measure whether they match the intended emotional tone.

Results

Model	Accuracy	F1-Score
Emotion Classifier	~39%	.2443
Response Generator	~33%	-

The classifier achieved an accuracy of 39.2% and a macro F1-score of 0.244. While these results indicate modest performance, they suggest that the model can onlearn meaningful emotional patterns beyond random guessing. Lower performance is expected due to the difficulty of fine-grained emotion recognition and the challenge of mapping labels between GoEmotions and DailyDialog, where some emotional categories overlap significantly.

The response generator achieved an accuracy of 33% during testing. Since the test included the use of the classifier for automation purposes, the accuracy of the response generator test is intertwined with that of the emotion classifier.



The confusion matrix for the emotion classifier test indicates overlap between various emotions, such as love and admiration and annoyance and anger. Due to the skew of the data set, a mass of neutral messages was tested compared to the other messages. There is a general trend of the classifier predicting the correct emotion, though the correlation on the heatmap is hard to see.

Discussion and Conclusion

This project explores emotionally aware dialogue generation through the combination of emotion classification and controllable text generation. By separating emotional understanding from response generation, the proposed system improves interpretability and control over conversational tone. Future work could include larger transformer architectures, finer-grained emotional conditioning, and human evaluation studies to further improve response quality and emotional realism. Furthermore, data sets with more intricate levels of emotion classification of messages would help. The mapping of GoEmotions' labels to Daily Dialogue categories proved to be more of a bottleneck than an innovation.

Works Cited

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